

An Explicit Infinite Family of Integer Solutions to $y^2 + x^3y + z^2 - 2 = 0$

Abstract

We prove, by an elementary Pell-type construction, that the Diophantine equation

$$y^2 + x^3y + z^2 - 2 = 0$$

has infinitely many integer solutions. The proof is completely explicit: from every positive term of the recursively defined Pell sequence

$$(c_0, d_0) = (1, 0), \quad c_{n+1} = 3c_n + 4d_n, \quad d_{n+1} = 2c_n + 3d_n,$$

we produce an integer solution (x_n, y_n, z_n) . The construction is self-contained and uses only elementary algebraic identities.

1 Statement of the result

We consider the integer equation

$$y^2 + x^3y + z^2 - 2 = 0. \tag{1.1}$$

The aim is not to classify all solutions, but to settle the requested alternative by proving that there are infinitely many integer solutions.

Theorem 1.1. *Equation (1.1) has infinitely many integer solutions. More precisely, define integer sequences $(c_n)_{n \geq 0}$ and $(d_n)_{n \geq 0}$ by*

$$(c_0, d_0) = (1, 0), \quad c_{n+1} = 3c_n + 4d_n, \quad d_{n+1} = 2c_n + 3d_n. \tag{1.2}$$

For every integer $n \geq 1$, put

$$\begin{aligned} a_n &= 2c_n^2 - 1, & R_n &= 2c_n(a_n - 1), & S_n &= a_n^2 - a_n - 1, \\ x_n &= 4c_nd_n, \\ z_n &= a_n(R_n - S_n), \\ y_n &= a_n(R_n + S_n) - \frac{x_n^3}{2}. \end{aligned} \tag{1.3}$$

Then $(x_n, y_n, z_n) \in \mathbb{Z}^3$ and

$$y_n^2 + x_n^3y_n + z_n^2 - 2 = 0.$$

Moreover, the integers x_n are strictly increasing for $n \geq 1$. Hence these solutions are pairwise distinct.

2 A square-completion reduction

The first step is to replace (1.1) by a representation problem of the shape $u^2 + 4z^2 = x^6 + 8$.

Lemma 2.1. *Let $x, y, z \in \mathbb{Z}$, and set*

$$u = 2y + x^3.$$

Then (x, y, z) solves (1.1) if and only if

$$u^2 + 4z^2 = x^6 + 8. \quad (2.1)$$

Conversely, if $x, u, z \in \mathbb{Z}$ satisfy (2.1) and

$$u \equiv x^3 \pmod{2}, \quad (2.2)$$

then

$$y = \frac{u - x^3}{2}$$

is an integer and (x, y, z) solves (1.1).

Proof. Starting from (1.1), we have

$$y^2 + x^3y + z^2 = 2.$$

Multiplying by 4 gives

$$4y^2 + 4x^3y + 4z^2 = 8.$$

Adding x^6 to both sides yields

$$x^6 + 4x^3y + 4y^2 + 4z^2 = x^6 + 8.$$

The left-hand side is exactly

$$(2y + x^3)^2 + 4z^2.$$

Thus (2.1) holds with $u = 2y + x^3$.

Conversely, assume $x, u, z \in \mathbb{Z}$ satisfy (2.1) and (2.2). Then $u - x^3$ is even, so

$$y = \frac{u - x^3}{2}$$

is an integer. Therefore $u = 2y + x^3$. Substituting this into (2.1) gives

$$(2y + x^3)^2 + 4z^2 = x^6 + 8.$$

Expanding the square and cancelling x^6 from both sides gives

$$4y^2 + 4x^3y + 4z^2 = 8.$$

Dividing by 4, we obtain

$$y^2 + x^3y + z^2 = 2,$$

which is equivalent to (1.1). □

3 An infinite Pell sequence

We shall need infinitely many integer pairs (c, d) satisfying $c^2 - 2d^2 = 1$. Instead of invoking Pell's theorem as a black box, we construct such pairs explicitly.

Lemma 3.1. *Let (c_n, d_n) be defined by (1.2). Then, for every $n \geq 0$,*

$$c_n^2 - 2d_n^2 = 1. \quad (3.1)$$

Furthermore, for every $n \geq 1$, both c_n and d_n are positive, and the sequences $(c_n)_{n \geq 0}$ and $(d_n)_{n \geq 0}$ are strictly increasing from $n = 0$ onward in the following sense:

$$c_{n+1} > c_n, \quad d_{n+1} > d_n \quad (n \geq 0).$$

In particular, there are infinitely many distinct positive integer solutions of $c^2 - 2d^2 = 1$.

Proof. First, $(c_0, d_0) = (1, 0)$, so

$$c_0^2 - 2d_0^2 = 1^2 - 2 \cdot 0^2 = 1.$$

Assume that $c_n^2 - 2d_n^2 = 1$. Using the recurrence (1.2), we compute

$$\begin{aligned} c_{n+1}^2 - 2d_{n+1}^2 &= (3c_n + 4d_n)^2 - 2(2c_n + 3d_n)^2 \\ &= (9c_n^2 + 24c_nd_n + 16d_n^2) - 2(4c_n^2 + 12c_nd_n + 9d_n^2) \\ &= 9c_n^2 + 24c_nd_n + 16d_n^2 - 8c_n^2 - 24c_nd_n - 18d_n^2 \\ &= c_n^2 - 2d_n^2 \\ &= 1. \end{aligned}$$

Thus (3.1) holds for every $n \geq 0$ by induction.

Next we prove positivity and strict growth. Since $(c_0, d_0) = (1, 0)$, we have $c_0 > 0$ and $d_0 \geq 0$. If $c_n > 0$ and $d_n \geq 0$, then

$$c_{n+1} = 3c_n + 4d_n > 0, \quad d_{n+1} = 2c_n + 3d_n > 0.$$

Hence $c_n > 0$ for all $n \geq 0$, and $d_n \geq 0$ for all $n \geq 0$.

Moreover,

$$c_{n+1} - c_n = 2c_n + 4d_n > 0$$

for every $n \geq 0$, because $c_n > 0$ and $d_n \geq 0$. Also

$$d_{n+1} - d_n = 2c_n + 2d_n > 0$$

for every $n \geq 0$, again because $c_n > 0$ and $d_n \geq 0$. Therefore (c_n) and (d_n) are strictly increasing in the stated sense. Since the pairs (c_n, d_n) are distinct and satisfy $c_n^2 - 2d_n^2 = 1$, this gives infinitely many positive integer solutions for $n \geq 1$. $\square$

Remark 3.2. Equivalently, (c_n, d_n) are the coefficients in

$$(3 + 2\sqrt{2})^n = c_n + d_n\sqrt{2}.$$

The recurrence proof above is included so that no external theorem on Pell equations is being used.

4 The auxiliary identities

The construction is based on forcing $x^6 + 8$ to be a sum of one square and four times another square. The following lemmas contain the required identities.

Lemma 4.1. *Let $c, d \in \mathbb{Z}$ satisfy*

$$c^2 - 2d^2 = 1. \quad (4.1)$$

Define

$$a = 2c^2 - 1, \quad b = 2cd. \quad (4.2)$$

Then

$$a^2 - 2b^2 = 1. \quad (4.3)$$

Proof. From (4.1), we have

$$2d^2 = c^2 - 1. \quad (4.4)$$

Using (4.2), we compute

$$\begin{aligned} a^2 - 2b^2 &= (2c^2 - 1)^2 - 2(2cd)^2 \\ &= 4c^4 - 4c^2 + 1 - 8c^2d^2. \end{aligned}$$

By (4.4),

$$8c^2d^2 = 4c^2(2d^2) = 4c^2(c^2 - 1) = 4c^4 - 4c^2.$$

Therefore

$$a^2 - 2b^2 = 4c^4 - 4c^2 + 1 - (4c^4 - 4c^2) = 1.$$

This proves (4.3). □

Lemma 4.2. *Let $c, d \in \mathbb{Z}$ satisfy $c^2 - 2d^2 = 1$, and define*

$$a = 2c^2 - 1, \quad R = 2c(a - 1), \quad S = a^2 - a - 1.$$

Then

$$R^2 + S^2 = a^4 - 3a^2 + 3. \quad (4.5)$$

Proof. Since $a = 2c^2 - 1$, we have

$$a + 1 = 2c^2. \quad (4.6)$$

Therefore

$$R^2 = (2c(a - 1))^2 = 4c^2(a - 1)^2 = 2(a + 1)(a - 1)^2, \quad (4.7)$$

where the last equality follows from (4.6). Expanding the right-hand side of (4.7),

$$\begin{aligned} 2(a + 1)(a - 1)^2 &= 2(a + 1)(a^2 - 2a + 1) \\ &= 2(a^3 - a^2 - a + 1) \\ &= 2a^3 - 2a^2 - 2a + 2. \end{aligned}$$

Also,

$$\begin{aligned} S^2 &= (a^2 - a - 1)^2 \\ &= a^4 - 2a^3 - a^2 + 2a + 1. \end{aligned}$$

Adding these two expansions gives

$$\begin{aligned} R^2 + S^2 &= (2a^3 - 2a^2 - 2a + 2) + (a^4 - 2a^3 - a^2 + 2a + 1) \\ &= a^4 - 3a^2 + 3. \end{aligned}$$

This is (4.5). □

Lemma 4.3. *Let $a, b \in \mathbb{Z}$ satisfy*

$$a^2 - 2b^2 = 1.$$

Set

$$x = 2b.$$

Then

$$x^2 + 2 = 2a^2, \tag{4.8}$$

and

$$x^4 - 2x^2 + 4 = 4(a^4 - 3a^2 + 3). \tag{4.9}$$

Consequently,

$$x^6 + 8 = 8a^2(a^4 - 3a^2 + 3). \tag{4.10}$$

Proof. Because $a^2 - 2b^2 = 1$, we have

$$2b^2 = a^2 - 1. \tag{4.11}$$

Since $x = 2b$,

$$x^2 + 2 = 4b^2 + 2 = 2(2b^2 + 1) = 2a^2,$$

which proves (4.8).

Next,

$$\begin{aligned} x^4 - 2x^2 + 4 &= (2b)^4 - 2(2b)^2 + 4 \\ &= 16b^4 - 8b^2 + 4 \\ &= 4(4b^4 - 2b^2 + 1). \end{aligned}$$

Using (4.11), we compute

$$\begin{aligned} 4b^4 - 2b^2 + 1 &= 4 \left(\frac{a^2 - 1}{2} \right)^2 - (a^2 - 1) + 1 \\ &= (a^2 - 1)^2 - a^2 + 2 \\ &= a^4 - 2a^2 + 1 - a^2 + 2 \\ &= a^4 - 3a^2 + 3. \end{aligned}$$

Therefore (4.9) follows.

Finally, the elementary factorization

$$x^6 + 8 = (x^2)^3 + 2^3 = (x^2 + 2)(x^4 - 2x^2 + 4)$$

together with (4.8) and (4.9) gives

$$x^6 + 8 = (2a^2) \cdot 4(a^4 - 3a^2 + 3) = 8a^2(a^4 - 3a^2 + 3).$$

This proves (4.10). □

5 Construction of the solutions

We now assemble the identities.

Lemma 5.1. *Let $c, d \in \mathbb{Z}$ satisfy*

$$c^2 - 2d^2 = 1.$$

Define

$$\begin{aligned} a &= 2c^2 - 1, & b &= 2cd, & x &= 2b = 4cd, \\ R &= 2c(a - 1), & S &= a^2 - a - 1, \\ u &= 2a(R + S), & z &= a(R - S). \end{aligned} \tag{5.1}$$

Then

$$u^2 + 4z^2 = x^6 + 8. \tag{5.2}$$

If, in addition, x is even, then

$$y = \frac{u - x^3}{2}$$

is an integer solution-producing value, namely (x, y, z) solves (1.1).

Proof. By Lemma 4.1, the integers a and b satisfy

$$a^2 - 2b^2 = 1.$$

Since $x = 2b$, Lemma 4.3 gives

$$x^6 + 8 = 8a^2(a^4 - 3a^2 + 3). \tag{5.3}$$

By Lemma 4.2,

$$R^2 + S^2 = a^4 - 3a^2 + 3.$$

Substituting this into (5.3), we get

$$x^6 + 8 = 8a^2(R^2 + S^2). \tag{5.4}$$

On the other hand, from the definitions of u and z ,

$$\begin{aligned} u^2 + 4z^2 &= (2a(R + S))^2 + 4(a(R - S))^2 \\ &= 4a^2(R + S)^2 + 4a^2(R - S)^2 \\ &= 4a^2((R + S)^2 + (R - S)^2). \end{aligned}$$

Now

$$(R + S)^2 + (R - S)^2 = (R^2 + 2RS + S^2) + (R^2 - 2RS + S^2) = 2R^2 + 2S^2 = 2(R^2 + S^2).$$

Hence

$$u^2 + 4z^2 = 4a^2 \cdot 2(R^2 + S^2) = 8a^2(R^2 + S^2).$$

Comparing this with (5.4), we obtain

$$u^2 + 4z^2 = x^6 + 8.$$

This proves (5.2).

If x is even, then x^3 is even. Also $u = 2a(R + S)$ is even. Therefore $u - x^3$ is even, and

$$y = \frac{u - x^3}{2}$$

is an integer. Lemma 2.1 then implies that (x, y, z) solves (1.1). □

Proof of Theorem 1.1. Let $n \geq 1$. By Lemma 3.1, the pair (c_n, d_n) satisfies

$$c_n^2 - 2d_n^2 = 1,$$

with $c_n, d_n \in \mathbb{Z}$ and $c_n, d_n > 0$. Apply Lemma 5.1 with $(c, d) = (c_n, d_n)$. The definitions in (5.1) become exactly the definitions in (1.3), with

$$u_n = 2a_n(R_n + S_n).$$

Because

$$x_n = 4c_n d_n,$$

the integer x_n is even. Therefore

$$y_n = \frac{u_n - x_n^3}{2} = a_n(R_n + S_n) - \frac{x_n^3}{2}$$

is an integer, and Lemma 5.1 gives

$$y_n^2 + x_n^3 y_n + z_n^2 - 2 = 0.$$

Thus every $n \geq 1$ gives an integer solution.

It remains only to prove that infinitely many distinct solutions are obtained. By Lemma 3.1, both (c_n) and (d_n) are strictly increasing and positive for $n \geq 1$. Hence, for every $n \geq 1$,

$$x_{n+1} = 4c_{n+1}d_{n+1} > 4c_n d_n = x_n.$$

So the values x_n are strictly increasing. In particular, the triples (x_n, y_n, z_n) are pairwise distinct. Therefore (1.1) has infinitely many integer solutions. $\square$

6 First examples

The first positive Pell pair produced by (1.2) is

$$(c_1, d_1) = (3, 2).$$

Then

$$a_1 = 2 \cdot 3^2 - 1 = 17, \quad R_1 = 2 \cdot 3 \cdot (17 - 1) = 96, \quad S_1 = 17^2 - 17 - 1 = 271,$$

and

$$x_1 = 4 \cdot 3 \cdot 2 = 24.$$

Therefore

$$z_1 = 17(96 - 271) = -2975,$$

and

$$y_1 = 17(96 + 271) - \frac{24^3}{2} = 17 \cdot 367 - 6912 = 6239 - 6912 = -673.$$

Thus

$$(x, y, z) = (24, -673, -2975)$$

is a solution. Since the equation depends on z only through z^2 , the sign-changed triple

$$(24, -673, 2975)$$

is also a solution.

For reference, the first few triples coming from the construction are listed below. In every row, changing the sign of z gives another solution.

n	c_n	d_n	x_n	(y_n, z_n)
1	3	2	24	$(-673, -2975)$
2	17	12	816	$(-68602753, -180466559)$
3	99	70	27720	$(-3043629943201, -7454236759199)$
4	577	408	941664	$(-121772121013218817, -294705964894072319)$

A direct verification of the first row is

$$(-673)^2 + 24^3(-673) + (-2975)^2 - 2 = 452929 - 9303552 + 8850625 - 2 = 0.$$

7 Conclusion

The recursive Pell sequence (1.2) gives infinitely many positive pairs (c_n, d_n) . The explicit formulas (1.3) then give infinitely many pairwise distinct integer triples (x_n, y_n, z_n) solving

$$y^2 + x^3y + z^2 - 2 = 0.$$

This proves, unconditionally and constructively, that the equation has infinitely many integer solutions. The construction is explicit and does not depend on any unproved classification theorem.